FSANet: Feature-and-Spatial-Aligned Network for Tiny Object Detection in Remote Sensing Images

Jixiang Wu, Zongxu Pan, *Member, IEEE*, Bin Lei, and Yuxin Hu

***Abstract*— Recently, many studies have successfully exploited convolutional neural networks to improve the performance of object detection in remote sensing images. However, detecting tiny objects is still challenging for two main neglected problems: i) The features of tiny objects are insufficient and prone to aliasing during the multiresolution aggregation process; ii) Tiny objects are position-sensitive, resulting in poor localization capabilities. In this article, we propose a feature-and-spatial aligned network (FSANet) to alleviate these issues. FSANet is an anchor-free detector that utilizes the alignment mechanism and progressive optimization strategy to obtain more discriminative features and accurate localization results. Concretely, we first present a feature-aware alignment module (FA^2M) to align features with different resolutions during fusion. By learning transformation offsets, FA^2M reencodes pixel-spatial information between feature maps of adjacent levels and adaptively adjusts the regular feature interpolation. Additionally, a spatial-aware guidance head (SAGH) is introduced to iteratively optimize network predictions in a coarse-to-fine fashion. SAGH first predicts the object shape at each spatial location on feature maps. For more precise predictions, it then captures geometry-aware convolutional features accordingly to update the coarse localization estimations. Extensive experiments are conducted on three tiny object detection datasets, i.e., AI-TOD, GF1-LRSD, and TinyPerson, demonstrating the effectiveness of our approach.**

***Index Terms*— Anchor-free method, convolutional neural network (CNN), feature alignment, geometry guidance, remote sensing, tiny object detection.**

I. Introduction

OBJECT detection is one of the fundamental tasks in the field of remote sensing image (RSI) interpretation. It plays an important role in various practical applications [1], [2], [3], [4], including intelligent monitoring, maritime management, precision agriculture, and urban planning. The aim of object detection in RSIs is to accurately locate and recognize highly structured targets in an image, such as ships and aircraft. Following the rapid development of Earth observation technologies, obtaining much data has become accessible and further promotes related scientific research.

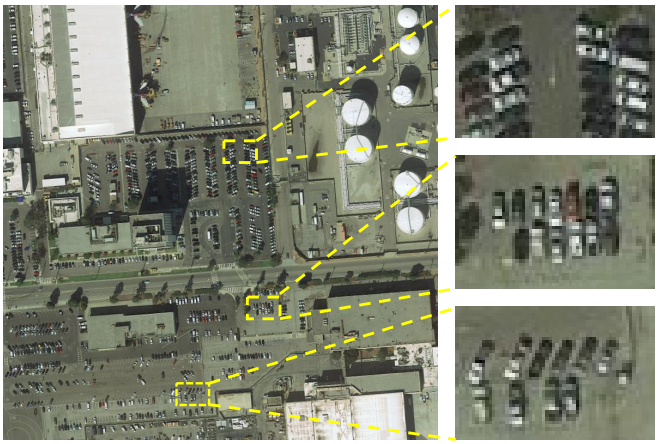


Fig. 1. Detecting tiny objects in RSIs. (Left) Original image sample with the size of 800 × 800. (Right) Zoomed in image slices for better observation. The tiny objects in RSIs are suffering challenges due to the characteristics of tiny size (smaller than 16 pixels), lacking abundant features, having a low SNR, etc.

Benefiting from the powerful feature representation capabilities of deep convolutional neural networks (CNNs) [5], [6], [7], deep learning-based methods are widely employed in various tasks to achieve higher performance. Furthermore, many state-of-the-art detection approaches [8], [9], [10], [11] in computer vision are extended to the remote sensing community. Objects in RSIs are characterized by dense arrangement, arbitrary orientation, etc. In practice, some researchers [12], [13], [14] have tried to generate rotation-invariant features or predict extra additional rotation parameters for representing rotated objects. However, previous methods mainly focus on solving the problem of objects with arbitrary orientations [15], [16], [17] in remote sensing object detection, and there remains an unsolved challenge caused by tiny objects. In general, objects whose absolute size is less than 16 × 16 pixels [18] are defined as tiny objects. As shown in Fig. 1, tiny objects in RSIs occupy only a few pixels and lack rich structural details, making it difficult to extract sufficient feature representations. Meanwhile, in real scenes, the background information is complicated and easily disturbed by illumination, weather, etc., resulting in a low signal-to-noise ratio (SNR) in RSIs. Thus, it is much more challenging to detect tiny objects than normal.

In the field of tiny object detection, CNN-based methods have gradually become mainstream. A straightforward way to

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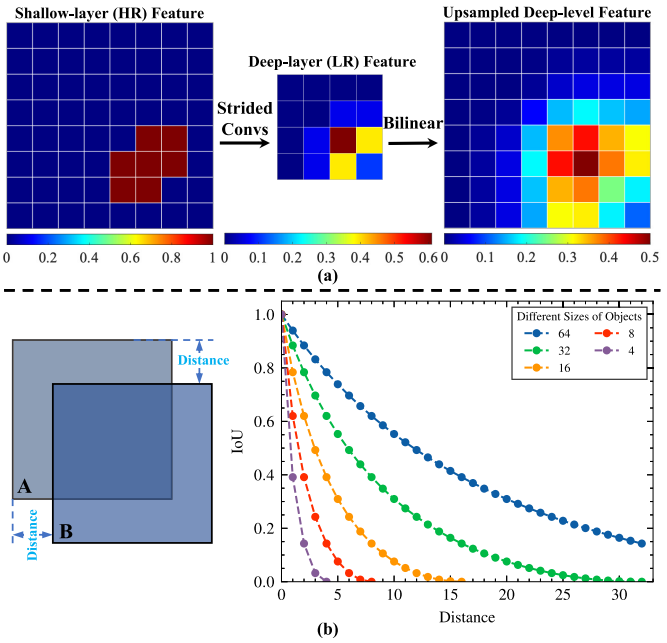


Fig. 2. Problem illustration. (a) Feature misalignment. CNNs use the pooling strategy or strided convolution to downsample the feature maps and extract deep semantic information. However, simply upsampling the deep features cannot recover the original spatial correspondence and will cause feature misalignment during fusing different levels of features. (HR: high-resolution; LR: low-resolution.) (b) Position sensitivity. We plotted the graph of IoU versus distance, where scatter curves of different colors represent objects of different sizes. As the distance between two bounding boxes increases, the IoU of tiny objects drops drastically, resulting in instability of the bounding box regression.

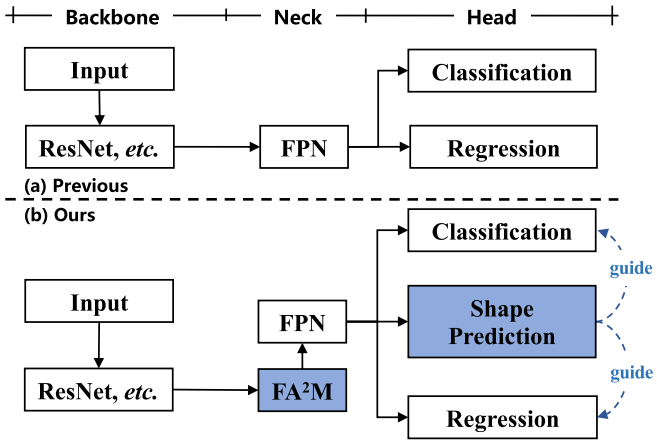


Fig. 3. Comparison of the previous work and our work. In the previous work, features extracted from backbone are directly upsampled and added to fuse each other, and the detection head is simple without considering object characteristics. In our work, we align different resolution features when constructing the FPN and utilize shapes of tiny objects to guide the feature refinement and update regression results in the detection heads.

improve the performance is to apply data augmentations, such as zooming in the image during data preprocessing [19] or oversampling the data that contains tiny objects to increase the number of objects [20], but it will bring extra training and inference time. Since the feature responses of tiny objects are very weak, further enhancement is required. For example, some works [21], [22] are devoted to fusing different levels of semantic information or expanding the network's receptive fields to obtain abundant contextual details. In addition, some researchers [23], [24] replace the standard Intersection-over-Union (IoU) metric used in anchor-based detectors [8], [9] with new label assignment criteria that allow the network to select enough positive samples for tiny object detection. However, we find that there are still two significant issues that fail to be considered and will be explained below.

On the one hand, feature misalignment occurs during the process of aggregating different resolution features. As we all know, modern detectors apply weight-shared convolutions to extract features from images and utilize multiple downsampling to deepen the network and construct the hierarchical structure that contains different semantic information. However, downsampling operations, (e.g., pooling or strided convolutions) will reduce the spatial resolution of feature maps and weaken the expression of spatial information. Especially for tiny objects, their features are hidden in the strong semantic information at the deep stages of the backbone. As mentioned above, feature pyramid network (FPN) [21] is an effective way to fuse semantic information at different levels and is widely used for tiny object detection. Specifically, low-resolution (LR) features are upsampled and forward-propagated to merge with adjacent high-resolution (HR) features through a top-down pathway. However, common interpolation operators (e.g., nearest neighbor, bilinear) only resample the feature map according to the regular mapping relationship of relative locations, ignoring the semantic correlation between pixels. As shown in Fig. 2(a), we visualize feature maps after different operations for comparison. It is observed that there are inaccurate correspondences between the upsampled and shallow-layer (HR) features. The feature representation of regions of interest is affected, and the background has negative responses. Therefore, feature misalignment occurs if the upsampled features are directly fused with the corresponding shallow features by element-wise addition. Aliased features will further impair the subsequent predictions. To tackle this issue, we propose a feature-aware alignment module (FA^{2}M) to learn spatial alignment information between adjacent-level features and adjust the interpolation process accordingly to generate aligned features.

On the other hand, tiny objects are position-sensitive. As illustrated in Fig. 2(b), we visualize the curve of IoUs between objects of different sizes as the distance increases. Compared to normal-sized objects (over 32 pixels in size), it is observed that tiny objects are more sensitive to positional changes. More specifically, a slight shift of the bounding box causes severe variation in IoU. For anchor-based detectors [8], [9], it is necessary to manually set the size and aspect ratio of the anchor boxes to extract candidate regions. However, under the strict IoU metric, it is difficult to match adequate positive samples for tiny objects, resulting in an imbalance between the foreground and background samples. Moreover, the general detection heads do not consider the special design for tiny objects, and locating tiny objects is more

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challenging. The poor regression capability further degrades detection performance. To tackle these issues, we adopt the anchor-free pipeline to build the detector for tiny object detection, which gets rid of the preset of anchor boxes and predicts the classification scores and regression bounding boxes in a per-pixel fashion. Besides, we propose a progressive regression mechanism to update the predicted bounding boxes iteratively.

Based on the above analyses, in this article, we propose the feature-and-spatial-aligned network (FSANet) for detecting tiny objects in RSIs. Compared with previous methods (see Fig. 3), our FSANet has the following advantages. First, FSANet is an anchor-free detector, which discards the design of anchor boxes and IoU metrics that would hamper the performance of tiny object detection. Especially, FSANet predicts the results at each location on feature maps and assigns the positive and negative samples according to whether the sample point is inside the ground truth box. Second, a FA^2M is proposed to mitigate the issue of feature misalignment during aggregating different resolution features. Inspired by the optical flow [25] which is used to estimate the motion of objects between consecutive frames, FA^2M learns the 2-D spatial transfer information between adjacent feature layers similarly. Then the transformation information is employed to adjust the interpolation operation for facilitating the feature alignment. Furthermore, FA^2M can be embedded into the FPN design to obtain more discriminative features. Third, in order to improve the localization capability of the network, we propose a spatial-aware guidance head (SAGH) that utilizes multiple stages to optimize the detection results. At the initial stage, the regression subnet predicts the coarse object shapes at each sampling location on feature maps. At the refinement stage, we exploit this geometric information and design a simple but effective shape-guided deformable convolution (SG-DCN) layer, with the aim to enhance the network perception of object shape and further generate more accurate predictions. This progressive optimization approach significantly improves the detection performance of tiny objects.

Our main contributions are summarized as follows.

1) We propose a FSANet designed for tiny object detection in RSIs, utilizing the alignment mechanism and progressive regression strategy to generate accurate predictions.
2) We design the FA^2M, which aligns the adjacent-level features during aggregation with the help of learnable spatial transformation information.
3) We present the SAGH that configures geometry-aware features and iteratively updates predicted results in a coarse-to-fine manner.
4) To demonstrate the effectiveness of our method, we conduct extensive experiments on three tiny object detection datasets, achieving state-of-the-art performance.

The rest of this article is organized as follows. We first briefly review related works in Section II. Then, the details of the proposed FSANet are illustrated in Section III. Next, we introduce the experimental settings, results and provide comprehensive ablation studies in Section IV. Finally, conclusions are summarized in Section V.

II. Related Work

A. Object Detection in RSIs

Various methods [14], [15], [16], [26], [27], [28], [29] have been proposed to facilitate the development of object detection in RSIs, including traditional methods and deep learning-based methods. The traditional approaches utilized handcrafted features or prior geographic information to match candidate regions, mainly focusing on single-class detection, such as ship detection [26], [30] and vehicles detection [27], [31]. For example, Zhu *et al.* [30] proposed an improved texture operator local multiple pattern (LMP) to extract ship features and used support vector machine (SVM) as the classifier to recognize true ships. Moranduzzo and Melgani *et al.* [31] and Chen *et al.* [32] used scale-invariant feature transform (SIFT) and histogram of oriented gradient (HOG) descriptors for feature extraction to detect vehicles, respectively. However, these methods only extract shallow features, having limitations in acquiring deep semantic information.

With the popularity of deep learning-based methods in computer vision [8], [9], [10], many works have made substantial progress in the field of object detection in RSIs. YOLT [33] extended YOLO [10] network from multiple perspectives to adapt to large-scene satellite imagery. Xia *et al.* [2] and Li *et al.* [3], respectively, built large-scale datasets with abundant annotations for object detection.

In order to solve the problem of rotation prediction, existing methods have tried to construct the rotated object detectors based on horizontal object detection schemes. S^2ANet [14] proposed a feature alignment module and an oriented detection module to realize aligned oriented features and consistent detections. R^3Det [16] designed cascaded regression subnets to locate objects accurately, and a feature refinement module was proposed to reencode the position information. Xie *et al.* [34] presented an effective two-stage oriented detector to generate high-quality oriented proposals.

What is more, some methods apply anchor-free models to represent objects in other forms. X-LineNet [35] proposed a bottom-up anchor-free detector, which represented aircraft as pairs of intersecting line segments. CFA [36] introduced convex-hull representation to alleviate the problem of feature aliasing and improve the performance of detecting irregular shape objects. Recently, few-shot learning [37], [38], weakly supervised learning [39], etc., are also introduced to the remote sensing community.

However, previous methods mainly focus on the challenges of arbitrary orientation, insufficient labeled instances, etc., while ignoring the problem of tiny object detection in real-world scenarios. Recently, researchers [18], [40] built datasets for tiny object detection in RSIs, providing new benchmarks for further studies. In this article, we propose an anchor-free detector for detecting tiny objects and conduct comprehensive experiments to validate the effectiveness of our method.

B. Tiny Object Detection

The performance of detecting tiny objects is far from satisfactory due to its tiny sizes and insufficient feature

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responses. In order to address this problem, different kinds of methods have been proposed. Briefly, these works can be categorized mainly into three aspects: data-based strategies, feature-enhanced learning, and sample assignment principles.

1) Data-Based Strategies: Generally, multiscale training [16], [19] and data augmentations like Mosaic [41] are typical methods to improve tiny object detection, but they will bring additional computation costs. SNIP [42] and SNIPER [43] proposed a training scheme that selectively train objects according to different scales of the image pyramid. Akyon *et al.* [19] proposed the slicing-aided inference and finetuning pipeline to augment the dataset by extracting and resizing patches from images. Kisantal *et al.* [20] oversampled images that contain small objects during training and used the "copy-paste" skill to enrich the number of small objects in images. Yu *et al.* [44] proposed the scale match approach to align the scale distribution of the training dataset with the pretraining dataset, which is beneficial for model transfer and fine-tuning.

2) Feature-Enhanced Learning: The tiny objects tend to occupy only a small portion of the image. Thus features of tiny objects are easily contaminated and ignored. Several works tried to obtain more feature representations for tiny objects. For example, SSD [45] predicted objects on multilayer feature maps, and higher resolution features are conducive to tiny object detection. FPN [21] utilized the top-down pathway and lateral connection to propagate and fuse different scale features. Later works, like PANet [46], BiFPN [47], and fusion factor [48], developed new ways of path aggregation for better feature fusion. Besides, background information provides additional supplements that help to recognize tiny objects. Therefore, dilated convolution [40], [49] and large-size kernels [22], [50] were adopted to obtain rich semantic information by enlarging receptive fields. TridentNet [51] proposed a parallel and weight-shared multibranch structure with different dilation rates to match the receptive fields of objects with different scales. Nonlocal [52] captured long-range dependencies and enhanced global context information representation. Other attention mechanisms [53], [54], [55] were also explored to strengthen features of tiny objects. In addition, researchers used generative adversarial network (GAN) [56], [57] to generate super-resolution features of tiny objects, making them similar to features of large-scale objects.

3) Sample Assignment Principles: The IoU metric is widely used in modern anchor-based detectors [8], [9], [10] to assign positive and negative samples during training. Nevertheless, tiny object detection requires careful manual setting of the sizes and aspect ratios of anchor boxes to ensure sufficient positive samples. ATSS [23] automatically divided samples according to the statistical characteristics of objects, which avoided additional hyperparameters. Meanwhile, tiny objects are sensitive to the change in position, where a slight shift may cause a significant disturbance of IoU. [24] utilized the normalized Wasserstein distance (NWD) to build detectors for tiny objects, including NWD-based label assignment, NWD-based NMS, and NWD-based loss. M-CenterNet [18] adopted the keypoint-based detector CenterNet [58] as baseline and proposed the design of multiple center points to improve the tiny object detection in RSIs.

Different from aforementioned methods, we find that the feature misalignment problem during multilevel feature fusion limits the feature representation of tiny objects. Thus, we propose an alignment mechanism, which learns pixel-spatial information between adjacent feature levels. For the problem that the network has poor abilities in locating tiny objects caused by position sensitivity, we introduce a progressive regression predictor, which first predicts coarse bounding boxes and utilizes geometry-aware features to update results.

III. Methodology

In this section, we first briefly introduce the overall architecture of the proposed FSANet. Then, we present the design of FA^{2}M that aligns the multiresolution features by employing transformation offset fields to interpolation. Next, we elaborate on the SAGH to mitigate the position sensitivity of tiny objects and improve the localization accuracy. Finally, we describe the process of network training and inference.

A. Overview

Fig. 4 illustrates the overall framework of the proposed FSANet, which adopts fully convolutional one-stage object detector (FCOS) [11] as the baseline. FSANet is an anchor-free detector that directly predicts the category and bounding box of objects at each location on feature maps instead of taking predefined anchor boxes as references. It consists of three components: a backbone for feature extraction, a feature pyramid network (FPN) [21] for feature fusion, and multiple parallel parameter-sharing detection heads for predicting object scores and locations. Specifically, We choose ResNet-50 [6] pretrained by ImageNet [7] as the backbone network in our design. Different from previous methods [8], [9], [11], we adopt the residual stages $\{C_3, C_4, C_5\}$ to construct the feature pyramid with levels $\{P_3, P_4, P_5\}$, removing useless levels $\{P_6, P_7\}$. On this basis, a FA^{2}M is embedded in FPN, which aims to capture abundant spatial transformation information between adjacent stages and align contextual features during aggregation. In the detection stage, we develop the SAGH, which predicts coarse boxes and estimates corresponding convolution offsets for all locations on the feature map. Finally, the regression branch generates box offsets to update the initially predicted bounding boxes, and the classification branch predicts object scores and center-ness.

B. Feature-Aware Alignment Module

Most modern object detection frameworks apply the multilevel structure to predict objects with the help of FPN [21]. The features with different semantic information are merged through lateral connections and top-down pathways. However, previous methods simply use the standard nearest neighbor sampling or bilinear interpolation to upsample the spatial resolution without considering the semantic alignment criterion of spatial locations between features, resulting in the problem of feature misalignment when propagating the semantically strong high-level features to lower-level features. Especially for tiny objects, the position information would drop rapidly as the network deepens, and more spatially specific information

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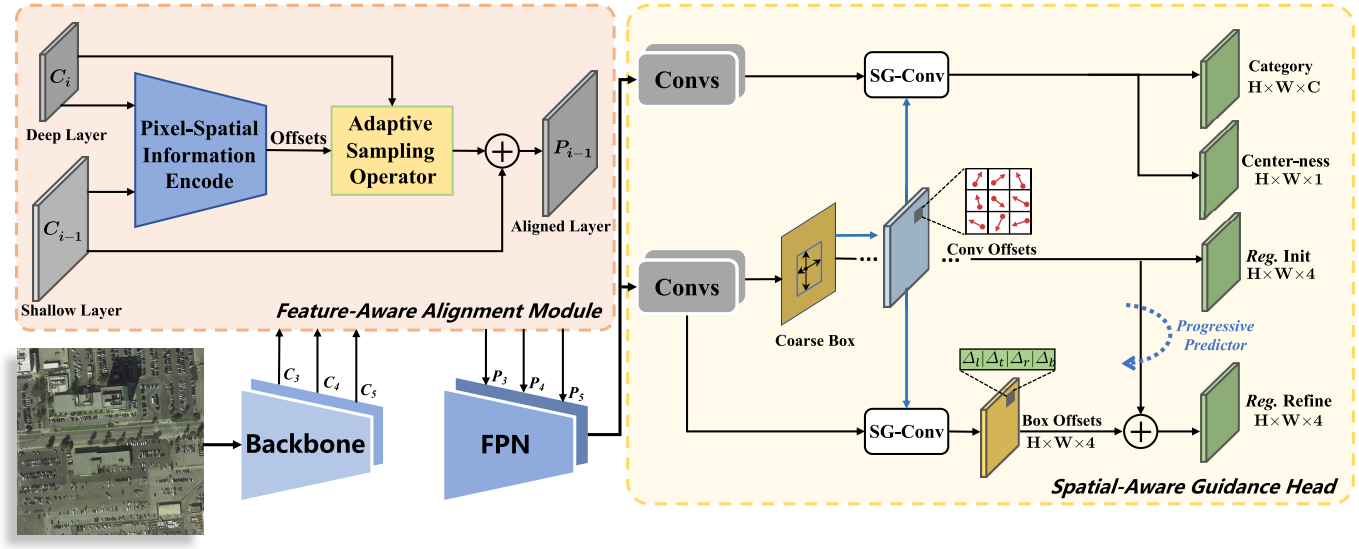


Fig. 4. Network architecture of our proposed FSANet for tiny object detection in RSIs. FSANet is an anchor-free detector equipped with the FA^{2}M and SAGH. Features from different stages extracted from the backbone network are fed into FA^{2}M to capture the pixel-spatial information between them and produce the aligned features during generating feature pyramid network. The shallow and deep features of the FA^{2}M input are adjacent layers, i.e., C_3 and C_4, C_4 and C_5. SAGH first predicts the coarse box and utilizes the geometry information to modulate convolutional features to further refine the detection results. The predicted shapes are under supervised. (SG-DCN: *Reg.* Init: the initial regression results; *Reg.* Refine: the refined regression results.)

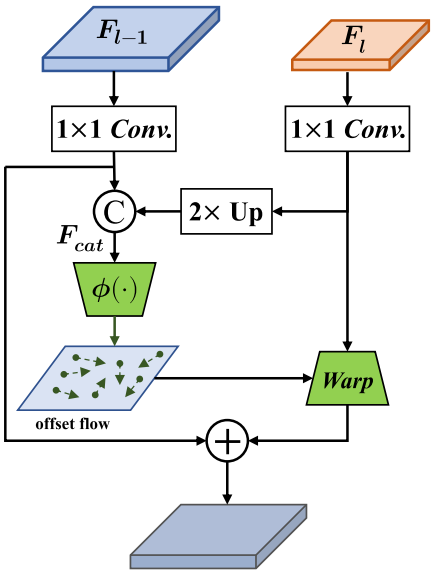


Fig. 5. Detailed structure of FA^{2}M, which is adopted to align the high-level feature map with its neighboring feature by generating an offset flow.

is required to recover features from low resolution to high resolution. Inspired by optical flow [25], we would like to obtain the relative transformation representation of pixels between adjacent feature layers to improve the alignment process, termed offset flow, similarly.

Based on the above observations, we propose a FA^{2}M to learn the implicit pixel-spatial transformation information and align features at different resolutions. The process illustration is shown in Fig. 5. Specifically, our FA^{2}M includes two main steps: i) Encode the pixel-spatial information and generate offset flow; ii) Adaptively resample and align features.

For the first step, given two feature maps $\mathbf{F}_l \in \mathbb{R}^{C_l \times H_l \times W_l}$ and $\mathbf{F}_{l-1} \in \mathbb{R}^{C_{l-1} \times H_{l-1} \times W_{l-1}}$ as inputs (where $\mathbf{F}_{l-1}$ and $\mathbf{F}_l$ are extracted from the backbone, representing high-resolution feature maps and low-resolution feature maps, and C, H, W, and l are the channel depth, height, width, and stage of the feature map, respectively.), we first apply two 1×1 convolutional layers followed by group normalization layers [59] to compress the dimension to the same size, e.g., $C = 256$. It is worth noting that the normalization operation is essential in order to keep the distributions of different-resolution features similar and avoid the influence of outliers. Then, we upsample the low-resolution feature by a factor of 2 via the standard bilinear interpolation to align the size and concatenate it with the high-resolution feature. The concatenated feature is denoted as $\mathbf{F}_{\text{cat}}$, formulated as follows:

$$\mathbf{F}_{\text{cat}} = \text{Cat}\left(\tilde{\mathbf{F}}_{l-1}, \tilde{\mathbf{F}}_l\right) \tag{1}$$

$$\left\{\tilde{\mathbf{F}}_{l-1}, \tilde{\mathbf{F}}_l\right\} = \{\text{Conv}_{1\times1}(\mathbf{F}_{l-1}), \text{Up}(\text{Conv}_{1\times1}(\mathbf{F}_l))\} \tag{2}$$

where $\tilde{\mathbf{F}}_{l-1}$ and $\tilde{\mathbf{F}}_l$ denote features that are obtained from $\mathbf{F}_{l-1}$ and $\mathbf{F}_l$ after channel compression and upsampling, respectively. $\text{Up}(\cdot)$ is bilinear interpolation operation, $\text{Conv}_{1\times1}$ is the 1×1 convolution layer, and $\text{Cat}(\cdot)$ is the concatenation operation. Considering that $\mathbf{F}_{\text{cat}}$ contains rich pixel context information, we design a lightweight offset subnetwork $\phi(\cdot)$ to generate the corresponding 2-D transformation offset flow. Concretely, we adopt a residual-like function to asymptotically estimate the expected outcome. We first create a reference field $\boldsymbol{P}_r \in \mathbb{R}^{2\times H_{l-1}\times W_{l-1}}$ of same spatial size as $\mathbf{F}_{\text{cat}}$, which is spatial pixel coordinates $\{(0,0), \ldots, (H_{l-1}-1, W_{l-1}-1)\}$. And then, the subnetwork $\phi(\cdot)$ is exploited to calculate the transformation offsets for each position on the reference field.

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Mathematically, the process is defined as follows:

$$\mathcal{O} = \boldsymbol{P}_r + \boldsymbol{P}_\Delta \tag{3}$$

$$\boldsymbol{P}_\Delta = \phi(\mathbf{F}_{\text{cat}}) \tag{4}$$

where $\mathcal{O} = \{\mathcal{O}_h, \mathcal{O}_w\}^{\text{T}} \in \mathbb{R}^{2\times H_{l-1}\times W_{l-1}}$ represents the output offset flow, showing the semantic modification direction between adjacent feature layers, and $\boldsymbol{P}_\Delta \in \mathbb{R}^{2\times H_{l-1}\times W_{l-1}}$ represents the offset field learned from $\mathbf{F}_{\text{cat}}$. For the implementation of the subnetwork, we employ a depth-wise convolution layer with the kernel size of 3×3 to capture local spatial information around each sampling position, and pass it through a point-wise convolution to generate the 2-D offset field $\boldsymbol{P}_\Delta$. Additionally, we modulate the amplitude of $\boldsymbol{P}_\Delta$ by a preset scalar s to stabilize the training, i.e., $\boldsymbol{P}_\Delta \leftarrow s \cdot \boldsymbol{P}_\Delta$. It is noted that depth-wise convolution is a grouped convolution strategy that can obtain abundant feature representations from different convolutional paths. Meanwhile, the 3×3 convolution has sufficient receptive fields to learn reasonable offsets, and we set `bais=False` in convolution settings to avoid overall position drift.

For the second step, we apply the differentiable bilinear interpolation [60] to resample the high-level feature guided by the offset flow $\mathcal{O}$. The algorithm samples linearly in the h and w direction around each calculated location, respectively, formulated in (5)

$$\mathcal{M}_l^{h,w} = \sum_{i=1}^{H_l}\sum_{j=1}^{W_l} \tilde{\mathbf{F}}_l^{i,j} \cdot \max(0, 1 - |\mathcal{O}_h - i|) \cdot \max(0, 1 - |\mathcal{O}_w - j|) \tag{5}$$

where (i, j) indexes each position on $\tilde{\mathbf{F}}_l$, and the $\mathcal{M}_l^{h,w}$ denotes the alignment output value at the location of (h, w). Then, we aggregate the aligned feature $\mathcal{M}_l$ with the high-resolution feature $\tilde{\mathbf{F}}_{l-1}$ by the element-wise addition.

Based on the aforementioned procedures, our FA^{2}M learns a reliable offset flow to represent the semantic transfer trend between adjacent feature layers, thereby alleviating the semantic gap problem during fusion and making features of tiny objects more discriminative. In addition, we embed FA^{2}M into the FPN design to build a more robust structure. The fused feature maps will be fed into the following detection subnets.

C. Spatial-Aware Guidance Head

As discussed previously, another challenging problem of tiny objects is position sensitivity, where a slight shift in position causes significant IoU score variation, as shown in Fig. 2(b). However, the conventional detectors [8], [9], [11] do not take into account the properties of tiny objects, having the following shortcomings. On the one hand, they employ standard convolutions to construct shared layers or parallel fully convolutional branches to extract features and predict classification probabilities and localization results, ignoring adapting features to objects of specific shapes and leading to suboptimal performance. On the other hand, especially for tiny objects, the anchor-based detectors confront the extreme imbalance between foreground and background due to the IoU-based assignment rule, leading to difficulties in network training and convergence.

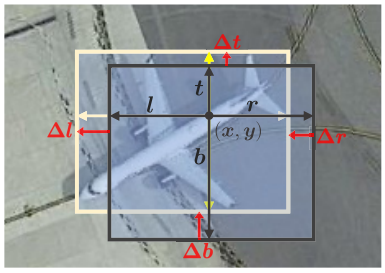


Fig. 6. Illustration of the bounding box prediction for SAGH. l, t, r, and b denote the distances from the location (x, y) to four sides of the box. Δl, Δt, Δr, and Δb are predicted box offsets generated in the refinement stage.

To address these issues, we propose a SAGH, which utilizes the geometric information of objects to dynamically configure the convolutional features and update the regression results in a coarse-to-fine manner, as shown in Fig. 4. Besides, SAGH is built based on the anchor-free paradigm, which avoids the operations on anchor boxes and predicts results in a per-pixel manner. The details are introduced in the following parts.

1) Progressive Predictor: The progressive learning algorithm is an effective strategy used in network refinement, so we integrate it into the network design. Taking the regression branch as an example, SAGH improves localization accuracy by iteratively optimizing predictions in multiple stages. According to FCOS [11], the regression subnet directly outputs a 4-D box vector (l, t, r, b) at each spatial location on feature maps to describe the shape of the predicted bounding box, which encodes the distances from the location to four sides (i.e., left, top, right, and bottom) of the bounding box. At the initial stage, SAGH first predicts a coarse box vector $V_p^i = (l_p^i, t_p^i, r_p^i, b_p^i)$ through three stacked 3×3 convolution layers. At the refinement stage, the well-designed shaped-guided deformable convolution is proposed to optimize the previous predictions, which generates the corresponding box offset vector $\Delta V = (\Delta l, \Delta t, \Delta r, \Delta b)$ at each location. Then, driven by the object localization loss, the network learns to update its own parameters and obtain more precise results, as shown in Fig. 6. Mathematically, the refined regression box vector V_p^{re} can be written as

$$V_p^{re} \leftarrow \left(l_p^i + \Delta l, t_p^i + \Delta t, r_p^i + \Delta r, b_p^i + \Delta b\right). \tag{6}$$

The vanilla convolution kernel samples and computes features over the feature map using a regular grid, but it is insufficient to model geometric transformations. Therefore, we introduce the SG-DCN to enhance the representation capacity of features, guided by the shape predicted by the network at each location, as shown in Fig. 7. The standard convolution operation is expressed as follows:

$$y(\boldsymbol{p}) = \sum_{\boldsymbol{p}_k \in \mathcal{R}} w(\boldsymbol{p}_k) \cdot x(\boldsymbol{p} + \boldsymbol{p}_k) \tag{7}$$

where x and y, respectively, denote the input and output feature map, w is the calculation weight for the summation, $\boldsymbol{p}$ is the sampling location, and $\mathcal{R}$ represents the regular grid that is instantiated as $\{(-1, -1), (-1, 0), \ldots, (+1, +1)\}$ for a convolution kernel of 3×3 and dilation of 1.

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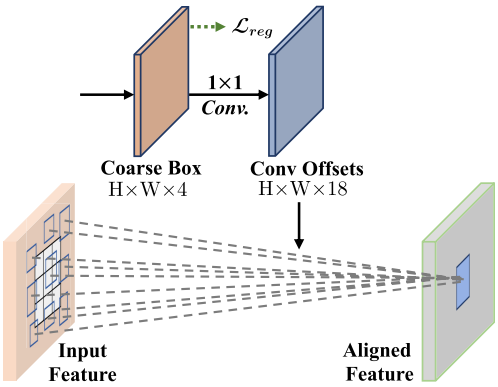


Fig. 7. Illustration of the SG-DCN, which utilizes predicted geometric information of objects to adjust sampling points of convolution kernels.

In particular, we employ the learnable offsets $\Delta \boldsymbol{p}_k$ to deform the regular grid into an irregular shape, which is transformed from the predicted bounding boxes at each location using a 1×1 convolution layer, as shown in Fig. 7. In this way, the convolution could flexibly adjust the receptive fields according to the estimated shape information and further improve the ability of decoding localization. The shaped-guided deformable convolution is formulated as follows:

$$y(\boldsymbol{p}) = \sum_{\boldsymbol{p}_k \in \mathcal{R}} w(\boldsymbol{p}_k) \cdot x(\boldsymbol{p} + \boldsymbol{p}_k + \Delta \boldsymbol{p}_k) \tag{8}$$

where $\Delta \boldsymbol{p}_k = \{(x_k, y_k) | k = 1, \ldots, N\}$, and $N = 9$.

Compared to the default deformable convolution [61], which uses its own features to learn convolutional offsets at each sampling position, our SG-DCN directly utilizes the predicted coarse boxes as references under strong supervision. In other words, our generated offsets are mapped from the geometric information, allowing for more accurate predictions and facilitating the network's convergence.

2) Gaussian Center-ness: According to FCOS [11], it designs an extra *"center-ness"* branch, in parallel with the classification branch to suppress the low-quality detections, as shown in Fig. 4. The center-ness represents the spatial distribution of object points, which is measured by the normalized distance from the current position to the center of the target. Given the regression target (l^*, t^*, r^*, b^*) for a prediction position, the cross-star center-ness (*C.C.*) used in [11] is defined as follows:

$$\text{C.C.}^* = \sqrt{\frac{\min(l^*, r^*)}{\max(l^*, t^*)} \times \frac{\min(t^*, b^*)}{\max(t^*, b^*)}}. \tag{9}$$

The weight distribution is visualized in Fig. 8(a), where the brighter color depicts the higher weight.

However, we find that C.C. is symmetrically distributed along the central axis (i.e., horizontal and vertical), and the weight decreases rapidly when the location is away from the central axis. This distribution focuses on the cross-star position and does not correspond to the real condition. Therefore, we propose the Gaussian center-ness (*G.C.*) to mitigate the above issues. Specifically, the Gaussian kernel is applied to compute the normalized Euclidean distance, expressed as follows:

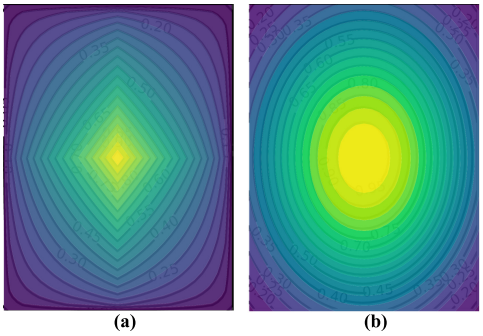


Fig. 8. Different forms of center-ness. (a) Cross-star center-ness used in FCOS [11] method. (b) Gaussian center-ness proposed in our work.

$$\text{G.C.}^* = \exp\left(-\left(\left(\frac{l^* - r^*}{l^* + r^*}\right)^2 + \left(\frac{t^* - b^*}{t^* + b^*}\right)^2\right)\right). \tag{10}$$

As shown in Fig. 8(b), the distribution spreads out from the object's center, and high-weight points are concentrated in the central region, which is different from C.C. Besides, the loss decreases steadily during training and achieves a better convergence performance in our experiments.

D. Training and Inference

1) Label Assignment: Our FSANet is an anchor-free detector that predicts a target bounding box and its category at each sampling location on feature maps and abandons the metric that defines samples by the IoU between prior anchor boxes and ground-truth boxes. For the regression branch, FSANet includes two stages to optimize the predictions. At the first stage, given the initial regression result $(l_p^i, t_p^i, r_p^i, b_p^i)$ at the location (x, y), the predicted box is considered positive if its corresponding sampling location is inside any ground-truth box, otherwise negative. Besides, if a location falls into multiple ground-truth boxes, the one with the smallest area is chosen as the regression target. Moreover, we specify the maximum regression range for each FPN level. If a prediction at level n satisfies $k_{n-1} \leq (l_p^i, t_p^i, r_p^i, b_p^i) \leq k_n$, it is reserved for further operations, otherwise ignored. The $\{k_2, k_3, k_4, k_5\}$ is set as $\{0, 64, 128, +\infty\}$, following [11]. At the refinement stage, for the updated bounding box $(l_p^{re}, t_p^{re}, r_p^{re}, b_p^{re})$ predicted at location (x, y), we calculate the center of the refined box $(x + (r_p^{re} - l_p^{re})/2, y + (b_p^{re} - t_p^{re})/2)$ as the "*pseudo*" location, which is utilized to reassign the label according to the same rules applied in the initial stage. For classification, the class label is determined in the second stage.

2) Loss Function: Here, we introduce loss functions used in FSANet to optimize each branch. More specifically, the classification loss is defined as follows:

$$\mathcal{L}_{\text{cls}} = \frac{1}{N_{\text{pos}}} \sum_{i=1}^{N} \text{FL}(\boldsymbol{p}_i, \boldsymbol{y}_i^*) \tag{11}$$

where N and N_{pos} denote the number of total samples and positive samples, respectively. $\boldsymbol{p}_i$ and $\boldsymbol{y}_i^*$, respectively, indicate the predicted probabilities and one-hot category labels of ith

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object. The focal loss [9] is adopted for classification to mitigate the imbalance of samples, expressed as follows:

$$\mathrm{FL}\left(p_i, y_i^*\right) = \begin{cases} -\alpha(1-p_i)^{\gamma}\log(p_i), & \text{if } y_i^* = 1 \\ -(1-\alpha)p_i^{\gamma}\log(1-p_i), & \text{if } y_i^* = 0 \end{cases} \tag{12}$$

where α and γ are two tunable hyperparameters, set as 0.25 and 2 by default [9], respectively.

The regression branch includes two stages to obtain precise bounding boxes, denoted as $\mathcal{L}_{\text{reg_init}}$ and $\mathcal{L}_{\text{reg_refine}}$

$$\mathcal{L}_{\text{reg}} = \frac{1}{N_{\text{pos}}}\sum_{i=1}^{N_{\text{pos}}}\mathcal{L}_{\text{IoU}}\left(\mathcal{B}_i, \mathcal{B}_i^*\right) \tag{13}$$

where $\mathcal{L}_{\text{IoU}}$ is IoU Loss [62], adopted for regression tasks. $\mathcal{B}_i$ and $\mathcal{B}_i^*$, respectively, denote the ith predicted bounding box and corresponding ground truth box. Only positive samples contribute to the final regression loss.

The center-ness branch is designed to estimate the importance of each prediction location. We apply the binary cross-entropy (BCE) Loss for computation

$$\mathcal{L}_{\text{ctr}} = \frac{1}{N_{\text{pos}}}\sum_{i=1}^{N_{\text{pos}}}\mathrm{BCE}\left(c_i, c_i^*\right) \tag{14}$$

where c_i and c_i^* denote the ith predicted result and ground-truth value, respectively.

Hence, our multitask loss can be written as follows:

$$\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{ctr}} + \lambda\mathcal{L}_{\text{reg_init}} + \mathcal{L}_{\text{reg_refine}} \tag{15}$$

where $\mathcal{L}_{\text{det}}$ is the combination of each component, and λ is a balance weight for $\mathcal{L}_{\text{reg_init}}$.

3) Inference: Our FSANet is an end-to-end detector. During inference, the network forward propagates the input image and obtains the classification probability p_i, center-ness c_i and the refined predicted bounding box $\mathcal{B}_i^r$ for each location on feature maps, respectively. We multiply the probability with corresponding center-ness as the ultimate score of object $s_i = p_i \cdot c_i$, following [11]. We retain results with scores higher than 0.05. Besides, the Soft-NMS [63] with a threshold of 0.5 is applied to filter out redundant detections.

IV. Experiments

In this section, we first describe the datasets used in this work, followed by the evaluation metrics and implementation details. We then conduct ablation studies and give detailed analyses to evaluate the effectiveness of each module. Finally, we report the performance of the proposed FSANet and compare it with other methods. Experimental results show our FSANet achieves new state-of-the-art performance for detecting tiny objects in RSIs.

A. Datasets

AI-TOD [18]: AI-TOD is specially built for tiny object detection in RSIs, based on five publicly available large-scale remote sensing datasets. It involves 28 036 images with 800 × 800 pixels and 700 621 labeled object instances in eight common categories. The mean size of AI-TOD is 12.8 pixels, and 85.6% of objects are smaller than 16 pixels, which is more suitable for tiny object detection tasks than other aerial datasets, such as dataset for DOTA [2] with the mean size of 55.3 pixels and 67.8% of objects larger than 16 pixels. Besides, 2/5 of the images are randomly split for training, 1/10 for validation, and 1/2 for testing.

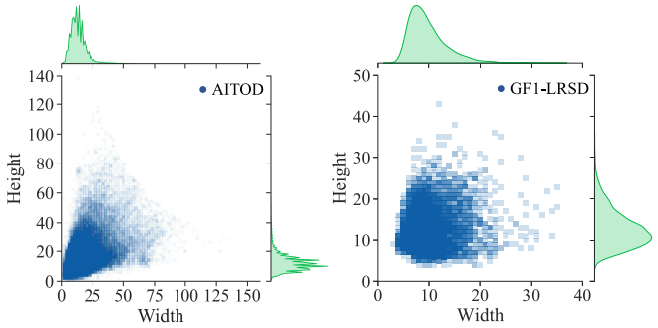


Fig. 9. Scale distributions of the different datasets. (Left) AI-TOD. (Right) GF1-LRSD.

GF1-LRSD [40]: GF1-LRSD is a challenging dataset for tiny ship detection in low-resolution RSIs. The original images are collected from Gaofen-1 satellite with a resolution of 16 m. After preprocessing, the dataset contains 4406 images with 512 × 512 pixels and 7172 instances in total. The mean size of the GF1-LRSD is 10.9 pixels and nearly 94% of objects are smaller than 16 pixels. For dataset splits, 3/5, 1/5, and 1/5 of the images are divided for training set, validation set, and testing set, respectively. In experiments, the training and validation sets are used for training and the rest for testing.

TinyPerson [44]: TinyPerson is another benchmark for tiny person detection in a long distance and with massive backgrounds. It contains a total of 1610 images collected from the Internet and 72 651 annotated instances with a mean size of 18.0 pixels. In experiments, 794 images of them are used for training, and the rest are used for inference.

We visualize the scale distributions of the above two datasets in Fig. 9, clearly showing that the scales of objects are mainly distributed in tiny sizes.

B. Evaluation Metrics

In experiments, the average precision (AP) is applied for the performance evaluation, defined as follows:

$$\mathrm{AP} = \int_0^1 \mathrm{P}(r)dr \tag{16}$$

where r is the recall rate, and $\mathrm{P}(r)$ is the corresponding precision. The recall and precision are calculated as follows:

$$\begin{aligned} \text{Recall} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \\ \text{Precision} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \end{aligned} \tag{17}$$

where TP, FP, and FN denote the numbers of true positive samples, false positive samples, and false negative samples defined under different IoU matching cases, respectively.

We apply the microsoft common objects in context (MS COCO) AP metric [64] for the evaluation of AI-TOD,

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TABLE I

ANALYSIS OF DIFFERENT FCOS ARCHITECTURE ON AI-TOD. "P.L." MEANS THE NUMBER OF FEATURE PYRAMID LEVELS. THE "MODIFIED-FCOS" IS CHOSEN AS BASELINE IN OUR WORK

Method	P.L.	AP	AP_{50}	AP_{75}	FLOPs	#Params
Vanilla-FCOS	P3∼P7	13.9	35.5	8.6	126.2G	31.9M
Modified-FCOS	P3∼P5	14.0	35.7	8.7	125.0G	30.7M
FSANet (Ours)	P3∼P5	16.3	41.4	9.8	125.2G	31.9M

TABLE II

ABLATION STUDIES OF FSANET VERSUS FCOS APPLYING DIFFERENT BACKBONES ON AI-TOD

Backbone	Method	AP	AP_{50}	AP_{75}
ResNet-50	Vanilla-FCOS	13.9	35.5	8.6
	FSANet (Ours)	16.3	41.4	9.8
ResNet-101	Vanilla-FCOS	14.4	36.5	8.6
	FSANet (Ours)	16.9	43.1	10.2
Swin-T	Vanilla-FCOS	17.9	44.1	11.3
	FSANet (Ours)	19.1	47.3	12.0

which reports the mean average precision (mAP) at IoU = 0.5:0.05:0.95. Specially, AI-TOD defines objects as *very tiny*, *tiny*, *small*, and *medium* with different size ranges: [2,8], [8,16], [16,32], and [32, $+\infty$]. GF1-LRSD is evaluated under the PASCAL VOC2012 metric [65] with an IoU threshold of 0.5. Following TinyPerson benchmark [44], objects of size [2,20] are defined as tiny and divided into three subintervals: $tiny1[2, 8]$, $tiny2[8, 12]$, and $tiny3[12, 20]$. The AP_{50} metric is used for performance evaluation.

C. Implementation Details

We implement FSANet based on Pytorch [66] and the `mmdetection` [67] toolkit. All models are trained on a single RTX 2080Ti graphics processing unit (GPU) with a batch size of 4. The ResNet-50 [6] pretrained on ImageNet [7] is adopted as the backbone. We use stochastic gradient descent (SGD) optimizer to optimize the training loss with a momentum of 0.9 and a weight decay of 1×10^{-4}. The hyperparameter λ is set to 0.2. On AI-TOD, the detector is trained for 12 epochs in total. The learning rate is initialized as 2×10^{-3} and divided by 10 at 8th and again at 11th epochs. On GF1-LRSD, the detector is trained for 24 epochs in total, and the initial learning rate is 2×10^{-3}, which is then divided by 10 at 16th and 22nd epochs. On TinyPerson, we follow settings in [44] to train and evaluate the detector. Due to GPU memory limitations, the original images need to be cropped into patches during training. The linear warm-up strategy [68] with a learning ratio of 0.333 is adopted at the first 500 iterations. Other settings are consistent with mmdetection unless otherwise specified.

D. Ablation Studies

In this section, we conduct comprehensive ablation experiments to validate the effectiveness of the proposed FSANet and different components, including FA^2M and SAGH. All results are reported on AI-TOD test set.

TABLE III

ABLATION STUDIES OF MODULES IN THE PROPOSED FSANET ON AI-TOD. "FA^2M": FEATURE-AWARE ALIGNMENT MODULE. "SAGH": SPATIAL-AWARE GUIDANCE HEAD

FA^2M	SAGH	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
		14.0	35.7	8.7	2.2	12.0	21.1	31.6
✓		14.9	38.1	8.8	2.5	13.0	21.8	32.9
	✓	15.6	39.6	9.3	3.4	13.9	21.8	32.9
✓	✓	16.3	41.4	9.8	4.4	14.6	23.4	33.3

TABLE IV

COMPARISONS OF DIFFERENT ALIGNING COMPONENTS ON AI-TOD. WE CALCULATE THE FLOATING-POINT OPERATIONS PER SECOND AND PARAMETERS OF DIFFERENT DESIGNS IN EXPERIMENTS

Method	AP	AP_{50}	AP_{75}	FLOPs	#Params
Nearest Neighbor	14.0	35.7	8.7	10.04G	2.69M
Bilinear Interpolation	14.3	36.3	8.8	10.04G	2.69M
Deconvlution	14.7	37.2	8.8	11.89G	3.87M
Offset Flow (Ours)	14.9	38.1	8.8	10.12G	2.70M

1) FCOS as Baseline: FCOS [11] detects different scale objects by assigning different FPN levels. In other words, the low levels are designed for small objects, and high levels are used to detect larger objects. For tiny object detection, we find that some structures are redundant. As shown in Table I, the vanilla-FCOS has a detection performance of 13.9% AP on AI-TOD test set. Then, we remove the P_6 and P_7 levels, named "Modified-FCOS". Modified-FCOS achieves similar results of 14.0% AP while slightly decreasing floating-point operations per seconds (FLOPs) (126.2 G→125.0 G) and parameters (31.9 M→30.7 M), which indicates that higher levels are unnecessary for our tasks.

Complexity: Modified-FCOS is chosen as the baseline for subsequent experiments. We propose FSANet to address existing problems of misalignment and position sensitivity. Compared with the baseline, our FSANet brings considerable improvements of 2.3% AP, 5.7% AP_{50}, and 1.2% AP_{75}. Meanwhile, our FSANet brings negligible additional computation, which is competitive with vanilla-FCOS in model size (31.9 M versus 31.9 M) and computation complexity (125.2 G versus 126.2 G). The results demonstrate the effectiveness of our method.

Different Backbones: We further conduct experiments to evaluate the performance of the proposed FSANet versus FCOS on different backbones. As shown in Table II, our FSANet significantly outperforms FCOS when applying different backbone networks. For instance, the detection AP yields a considerable margin of 2.5% AP (14.4% versus 16.9%) by employing ResNet-101. When integrating the latest Swin-T [69] into the network, our FSANet still outperforms FCOS with 1.2% AP improvements, demonstrating its remarkable robustness and effectiveness. In addition, it can be found that with the enhancement of the backbone network feature extraction ability, the tiny object detection performance is also improved gradually, e.g., AP: 16.3%→16.9%→19.1%,

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TABLE V

COMPARISONS OF FA^{2}M WITH OTHER SIMILAR METHODS ON AI-TOD. WE CALCULATE THE FLOPS AND PARAMETERS OF THESE MODULES IN EXPERIMENTS

Method	AP	AP_{50}	AP_{75}	FLOPs	#Params
Flow in [70]	14.4	36.6	8.5	10.16G	2.71M
FaPN in [71]	14.8	37.7	8.9	25.28G	6.44M
FA^{2}M (ours)	14.9	38.1	8.8	10.12G	2.70M

TABLE VI

ABLATION STUDIES OF THE SUBNETWORK $\phi(\cdot)$ WITH DIFFERENT KERNEL SIZES IN THE FA^{2}M DESIGN ON AI-TOD

Kernel Size	AP	AP_{50}	AP_{75}
1×1	15.8	40.4	9.5
3×3	16.3	41.4	9.8
5×5	16.0	41.0	9.3
7×7	15.9	40.5	9.3

TABLE VII

DETAILED ABLATIONS OF SAGH ON AI-TOD. G.C. AND P.P. DENOTE GAUSSIAN CENTER-NESS AND PROGRESSIVE PREDICTOR, RESPECTIVELY

G.C.	P.P.	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
-	-	14.0	35.7	8.7	2.2	12.0	21.1	31.6
✓	-	14.7	37.2	9.2	2.3	12.8	21.4	32.8
✓	✓	15.6	39.6	9.3	3.4	13.9	21.8	32.9

which verifies the importance of the well-designed feature representation.

2) Effectiveness of FA^{2}M: We first investigate the effectiveness of the proposed FA^{2}M, which predicts the offset flow to align different resolution features during aggregation. As reported in Table III, by embedding FA^{2}M into the baseline, the detection AP improves from 14.0% to 14.9%. The AP_{50} is improved by 2.4% (35.7%→38.1%). The AP_t is improved by 1.0% (12.0%→13.0%). For objects of different sizes, the performances are also improved in different proportions. This proves that the semantic information misalignment during feature fusion would degrade detection performance. Our FA^{2}M learns relative semantical offsets between two feature maps and modifies the low-resolution features during upsampling to obtain consistent feature representations at each spatial location with high-resolution features.

Different Aligning Components: We further conduct experiments by replacing the offset flow in FA^{2}M with other alignment components to verify the effectiveness of the proposed method. As shown in Table IV, FA^{2}M with offset flow consistently outperforms other methods, reaching the detection AP of 14.9%. The nearest neighbor and bilinear interpolation only upsample to align feature maps size, and the pixel correlation between different resolution features is ignored. Thus, these two components obtain inferior performances, e.g., the result of bilinear has a gap of 0.6% AP compared to our FA^{2}M. The deconvolution layers are

TABLE VIII

COMPARISONS OF DIFFERENT TYPES OF DEFORMABLE CONVOLUTIONS ON AI-TOD

Method	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
w/ DCN	15.9	40.7	9.5	3.6	14.3	22.2	32.3
w/ SG-DCN	16.3	41.4	9.8	4.4	14.6	23.4	33.3

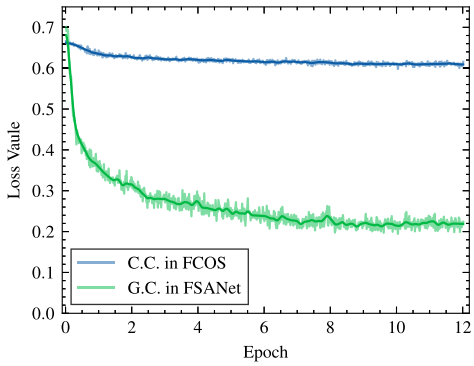


Fig. 10. Loss curves of different center-ness. The cross-star center-ness decreases slowly and has difficulties in convergence during training. The Gaussian center-ness converges to a lower level as training progresses.

operated on low-resolution features to produce corresponding high-resolution features. However, the deconvolution depends on extra learnable parameters to increase the ability of the network and does not utilize the semantical information of the adjacent feature map as guidance. Different from them, our FA^{2}M first aligns the spatial size of adjacent feature maps, then learns transformation information from the concatenated features to adjust interpolation location using differentiable bilinear interpolation [60]. In this way, the high-resolution features contribute to the final offset flow generation, and FA^{2}M takes advantage of contextual and spatial information, achieving the best performance. The AP_{50} is also improved by 2.4 points over baseline. In addition, we calculate the parameters and FLOPs of these methods. It can be found that our proposed FA^{2}M brings negligible computational complexity (only 0.08 G FLOPs and 0.01 M Params) compared to the baseline, further indicating the efficiency of the proposed scheme.

Comparisons of FA^{2}M With Other Similar Methods: Here we further examine the design of our FA^{2}M with other solutions. As shown in Table V, our FA^{2}M achieves better performance with fewer parameters and FLOPs. Flow in [70] is used to obtain precise object boundaries in semantic segmentation tasks. However, it ignores the limited feature representation of tiny objects. For feature-aligned pyramid network (FaPN) in [71], which adopts channel attention mechanism [53] and deformable convolutional network (DCN) [61] to select and align features in FPN design, it is not designed for tiny object detection and relies on additional learned parameters to improve the network. Unlike them, our FA^{2}M focuses on the alignment and representation of tiny object features. Specifically, we first normalize the different resolution features that can reduce the impact of data distribution differences and promote network learning. Then, we combine depth-wise

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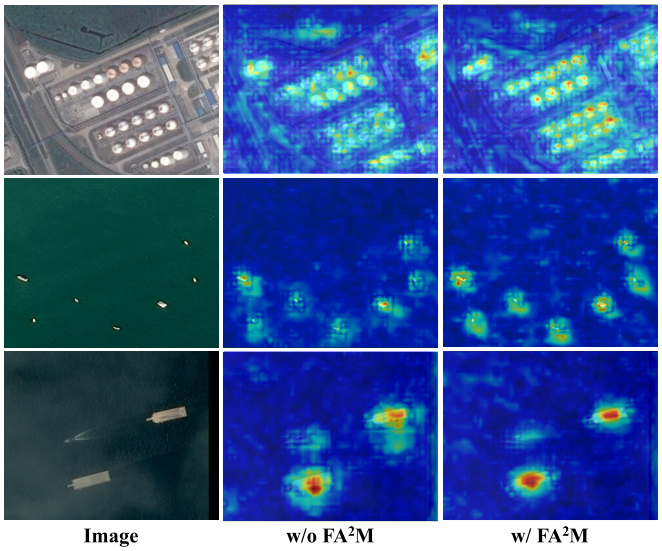


Fig. 11. Visualization of heatmaps under different settings. The heatmaps are extracted from P_3 level of FPN. (First Column) Input images. (Second Column) Heatmaps w/o FA^{2}M. (Third Column) Heatmaps w/ FA^{2}M.

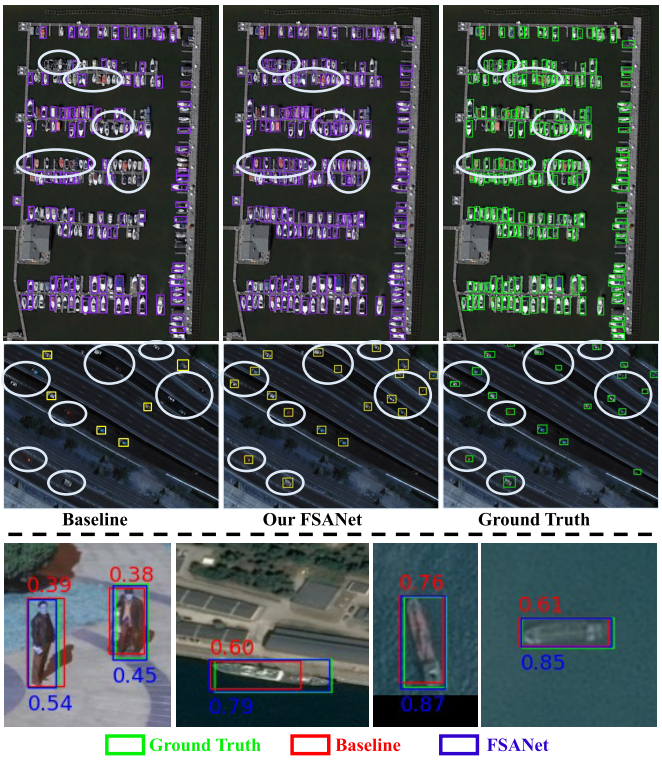


Fig. 12. Visualization of detection results by baseline and our FSANet. **Top** Separate results to inspect the overall performance. The oval circles are used to highlight differences. From left to right: baseline results, FSANet results, and ground truth. **Bottom** Merged and zoomed-in results for detailed comparisons. Numbers represent predicted scores. Colors represent different methods. (Green) Ground Truth. (Red) Baseline. (Blue) FSANet.

and point-wise convolutions to generate 2-D offset field. The depth-wise Conv extracts independent spatial dimension information in each channel and avoids different channel information coupling, which can learn more effective spatial transformation information for feature interpolation and reduce the network FLOPs. Thus, FA^{2}M yields 0.5% AP and 0.4% AP_{50} improvements over the methods in [70] and [71], respectively. Meanwhile, our method has 15.16 G FLOPs and 3.74 M Params less than FaPN [71], but better performance.

On Kernel Size: We also explore the impact of the subnetwork $\phi(\cdot)$ applying different convolutional kernel sizes in the FA^{2}M design, which is adopted to extract transformation offsets at each sampling position. The kernel sizes are set to 1×1, 3×3, 5×5, and 7×7, respectively. The experimental results are listed in Table VI. It can be observed that a 3×3 depth-wise convolution layer is sufficient for the network, achieving the best performance of 16.3% AP. We find that increasing the kernel size would not bring additional improvements, and the performance drops slightly, e.g., AP: 16.3%→16.0%→15.9%. We speculate that the reason is that larger kernel sizes (i.e., larger receptive fields) would introduce irrelevant information to interfere with offset generation.

3) Effectiveness of SAGH: As described above, the proposed SAGH predicts bounding boxes in a coarse-to-fine manner, and we conduct ablation experiments to validate the effectiveness of SAGH. The quantitative results are shown in Table III. By applying SAGH to the baseline, it leads to considerable performance gains in all AP metrics on AI-TOD over the baseline, as depicted in 1st and 3rd rows in Table III. For instance, applying SAGH improves the AP score by 1.6% (14.0% versus 15.6%). For different size objects, the performance is, respectively, boosted by 1.2% AP_{vt} (2.2% versus 3.4%) for *very tiny* objects, and 1.9% AP_t (12.0% versus 13.9%) for *tiny* objects, which demonstrates the capability of our SAGH for detecting tiny objects. In particular, SAGH first predicts coarse bounding boxes at each sampling location and generates the corresponding box offsets guided by the geometric information of objects to update the final results iteratively.

On Gaussian Center-ness and Progressive Predictor. SAGH includes two components in the network design: the Gaussian center-ness (G.C.) and the progressive predictor (P.P.). In order to further understand the impact of each part, we perform extra experiments, and the results are shown in Table VII. Replacing C.C. with G.C. yields a performance improvement of 0.7% AP (14.0% versus 14.7%) and 0.8% AP_t (12.0% versus 12.8%). The center-ness defines a distribution to assign weights to each prediction location, thereby reducing low-quality predictions. We use the normalized Gaussian Kernel function for implementation, which is more in line with the actual situation. We visualize different center-ness loss curves in Fig. 10, which indicates that our method has better convergence. In addition, we can observe that employing P.P. (the 2nd versus 3rd row in Table VII) further improves the performance, especially for tiny objects. The performance of AP_{vt} and AP_t are boosted by 1.1% and 1.1%, respectively, indicating the effectiveness and practicality of our method. Compared to the original design in FCOS [11] that uses four standard 3×3 convolutions to decode the bounding boxes, our progressive predictor builds an additional offset prediction part, which

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can generate compensating offsets at each sampling location based on the initially predicted bounding boxes, and further optimize predictions in a coarse-to-fine manner.

On Deformable Convolution. As described in Sec III-C, we present the SG-DCN to enhance the network's capability to represent geometric information. Our SG-DCN generates the geometrically dependent convolution offsets from the initially predicted bounding boxes (supervised by the regression loss) at each sampling location. As can be noticed in Table VIII, the proposed scheme produces better results compared to the standard deformable convolution, e.g., SG-DCN achieves 0.4% AP and 0.7% AP_{50} improvement over DCN in our experiment, indicating the superiority of our method.

4) Effectiveness of Combining FA^2.M and SAGH: With the joint addition of FA^{2}M and SAGH into the baseline, the performance compared to the models with either FA^{2}M or SAGH incorporated can be further enhanced. As shown in Table III, our FSANet outperforms the baseline by a large margin in all AP metrics, e.g., the detection AP obtains a noticeable increment of 2.3% (14.0% versus 16.3%). This phenomenon proves that our proposed FA^{2}M and SAGH are complementary. On the one hand, our FA^{2}M could help mitigate the aliasing effect of different resolution features fusion. The background noises are suppressed, and features around targets are enhanced. On the other hand, SAGH utilizes the geometric information of objects to configure convolutional features and further refine the regression and classification results in a cascaded fashion.

E. Visualization Analysis

1) Feature Alignment: As shown in Fig. 11, we visualize the heatmaps with and without FA^{2}M to better analyze and understand the benefits of our approach. We extract the feature maps at P_3 level of FPN and then apply the `maxpooling` operation on the channel dimension for compression. It is manifest that FA^{2}M can highlight the right objects and aggregate features to the corresponding locations of targets. Besides, features of tiny objects are enhanced and easily captured when equipped with FA^{2}M (the second row in Fig. 11). The main reason lies in the fact that simply adding different resolution features together does not work well due to misalignment, and our FA^{2}M learns the offset flow and aligns features to facilitate the feature representation of objects.

2) Tiny Object Detection: In Fig. 12, the detection results over the score threshold of 0.3 produced by FSANet and baseline are shown for clear comparison. It can be seen that our FSANet has fewer missed detections and achieves better performance. For dense scenes, our method also predicts accurate bounding boxes. What is more, we visualize the merged and zoomed-in results in the lower part of Fig. 12 to better analyze the proposed approach, where the red and blue colors, respectively, denote results predicted by baseline and FSANet. We can observe that our method produces satisfactory prediction results, achieving not only more accurate localization but also higher prediction scores. The results demonstrate that the proposed FSANet can generate more discriminative representations and progressively learn regression offsets to optimize prediction boxes.

F. Comparisons With Other Approaches

In this section, we compare our proposed FSANet with other previous methods on three tiny object detection datasets: AI-TOD [18], GF1-LRSD [40], and TinyPerson [44]. Table IX, X, and XI detail the quantitative detection results on different datasets. Figs. 13–15 show the visualization results of the proposed method.

1) Results on AI-TOD: As depicted in Table IX, our FSANet is compared with other methods on AI-TOD test set, including anchor-based detectors (e.g., Faster-RCNN [8], RetinaNet [9], and YOLOv3 [10]) and anchor-free detectors (FCOS [11], RepPoints [75], CornerNet [76], and CenterNet [58]). It is noted that results of other methods are referred from [18]. Obviously, our FSANet achieves the performance of 16.3% AP with ResNet-50 backbone and $1\times$ training times and exceeds others by a large margin. When applying longer training epochs and larger backbone models, the performance of FSANet can be further improved. We establish the new state-of-the-art 22.6% AP with a transformer-based (i.e., Swin-T) backbone and total of 24 training epochs.

As an anchor-free detector, our FSANet outperforms M-CenterNet [18] by 1.8% AP with ResNet-50 backbone. The AP_{75} is correspondingly improved from 6.4% to 9.8%, which indicates that our method has a better capability in high-quality localization. Likewise, the proposed FSANet also yields sizeable margins over other anchor-free detection paradigms. For instance, our method, respectively, surpasses RepPoints [75] by 7.1% AP and CornerNet [76] by 7.3% AP with ResNet-50 backbone. Consistent gains are also observed across other types of AP metrics. For CornerNet [76], it is designed to predict and group paired corners to generate bounding boxes. However, it can easily produce incorrect clustering results, especially in dense scenes with many objects of the same category. RepPoints [75], which aims to learn a set of adaptive sample points to represent the objects, is a complex task for detecting tiny objects. Because the feature patterns of tiny objects are limited and difficult to extract. Instead, our FSANet regresses bounding boxes at each sampling position in a dense prediction manner, which can reduce the difficulty of network optimization, while the well-designed FA^{2}M and SAGH can further align features and improve localization accuracy. Meanwhile, the proposed FSANet shows superior improvements in detecting tiny objects over anchor-based methods. Compared to Faster-RCNN [8], the AP_{vt} and AP_t are boosted by 4.4 and 6.3 points (0.0% versus 4.4% and 8.3% versus 14.6%), respectively. Other methods also show weaker performance, similarly. This is mainly because the IoU-based matching criterion will cause the targets to fail to be assigned to appropriate anchor boxes and exacerbate the imbalance of positive and negative samples.

Moreover, we report the class-wise results with $AP_{50:95}$ metric [64] in Table IX. It is worth noting that our FSANet achieves best detecting performances in 7/8 categories, especially for some hard tasks (e.g., bridge, wind-mill, and

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TABLE IX

COMPARISONS OF THE PROPOSED FSANET WITH PREVIOUS STATE-OF-THE-ART METHODS ON AI-TOD. WE REPORT THE CLASS-WISE DETECTION RESULTS WITH $AP_{50:95}$. THE CATEGORY NAMES ARE ABBREVIATED AS FOLLOWS: AI-AIRPLANE, BR-BRIDGE, ST-STORAGE TANK, SH-SHIP, SP-SWIMMING POOL, VE-VEHICLE, PE-PERSON, WM-WIND MILL. $(\cdot)^{\dagger}$ MEANS $2\times$ TRAINING TIMES

Method	Backbone	AI	BR	ST	SH	SP	VE	PE	WM	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
Anchor-based Method																
Faster R-CNN [8]	ResNet-50	22.71	3.87	20.18	19.02	8.90	11.88	4.49	0.32	11.4	27.0	8.0	0.0	8.3	23.1	24.5
Cascade R-CNN [72]	ResNet-50	25.57	7.47	23.33	23.55	10.81	14.09	5.34	0.00	13.8	30.8	10.5	0.0	10.6	25.5	26.6
YOLOv3 [10]	DarkNet-53	7.14	2.60	3.66	10.69	0.61	8.50	2.13	0.40	4.5	14.2	1.7	2.1	4.6	5.9	6.2
YOLOv5 [73]	CSP-DarkNet	9.29	6.37	20.28	32.86	0.86	19.28	5.28	0.70	11.9	28.5	7.6	3.1	12.0	14.2	20.3
RetinaNet [9]	ResNet-50	0.01	6.62	1.84	20.87	0.06	5.67	1.75	0.53	4.7	13.6	2.1	2.0	5.4	6.3	7.6
SSD-512 [45]	VGG-16	14.52	3.13	10.89	13.05	1.92	7.84	3.12	1.48	7.0	21.7	2.8	1.0	5.4	11.5	13.5
TridentNet [51]	ResNet-50	9.67	0.77	12.28	17.11	3.20	11.87	3.98	0.94	7.5	20.9	3.6	1.0	5.8	12.6	14.0
Anchor-free Method																
FCOS [11]	ResNet-50	19.28	7.37	26.05	27.40	5.77	17.37	5.63	2.56	13.9	35.5	8.6	2.7	12.0	20.2	32.2
FoveaBox [74]	ResNet-50	13.75	0.00	18.51	17.70	0.03	11.42	3.38	0.00	8.1	19.8	5.1	0.9	5.8	13.4	15.9
RepPoints [75]	ResNet-50	2.92	2.34	21.37	26.40	0.00	15.16	5.39	0.00	9.2	23.6	5.3	2.5	9.2	12.9	14.4
CornerNet [76]	Hourglass-104	10.63	11.81	14.05	16.94	4.41	6.96	4.82	2.67	9.0	25.4	4.3	2.9	12.9	11.5	7.0
CenterNet [58]	DLA-34	17.43	9.46	25.93	21.86	6.21	16.54	8.12	1.94	13.4	39.2	5.0	3.8	12.1	17.7	18.9
M-CenterNet [18]	DLA-34	18.59	10.58	27.55	22.27	7.53	18.60	**9.17**	2.03	14.5	40.7	6.4	6.1	15.0	19.4	20.4
FSANet (Ours)	ResNet-50	22.21	10.81	28.24	29.84	9.40	19.57	5.89	4.23	16.3	41.4	9.8	4.4	14.6	23.4	33.3
FSANet† (Ours)	ResNet-50	27.86	14.35	31.33	38.62	14.78	22.03	7.48	**5.94**	20.3	48.1	14.0	6.3	19.0	26.8	36.7
FSANet† (Ours)	ResNet-101	27.77	14.83	32.39	**41.59**	15.21	22.59	8.27	5.90	21.1	49.6	14.6	**7.5**	19.9	28.5	35.7
FSANet† (Ours)	Swin-T	**30.90**	**15.13**	**35.00**	40.32	**19.81**	**24.93**	8.92	5.58	**22.6**	**52.8**	**15.6**	7.4	**21.6**	**29.1**	**38.5**

TABLE X

COMPARISONS OF THE PROPOSED FSANET WITH PREVIOUS STATE-OF-THE-ART METHODS ON GF1-LRSD. THE RESULTS IS EVALUATED UNDER PASCAL VOC2012 METRICS

Method	Backbone	mAP
Faster-RCNN [8]	ResNet-50	60.49
RetinaNet [9]	ResNet-50	79.00
SSD-512 [45]	VGG-16	72.28
YOLOv3 [10]	DarkNet-53	67.82
YOLOv5 [73]	CSP-DarkNet	80.52
FCOS [11]	ResNet-50	74.16
CornerNet [76]	Hourglass-104	78.10
ATSS [23]	ResNet-50	73.84
SCRDet [22]	ResNet-50	69.29
R^3Det [16]	ResNet-50	73.04
LR-TSDet [40]	ResNet-50	83.87
FSANet (Ours)	ResNet-50	87.41

TABLE XI

COMPARISONS OF THE PROPOSED FSANET WITH PREVIOUS METHODS ON TINYPERSON. RESULTS ARE EVALUATED UNDER AP_{50} METRICS

Method	AP_{50}^{tiny}	AP_{50}^{tiny1}	AP_{50}^{tiny2}	AP_{50}^{tiny3}
RetinaNet [9]	33.53	12.24	38.79	47.38
Faster RCNN [8]	47.35	30.25	51.58	58.95
Grid RCNN [77]	47.14	30.65	52.21	57.21
Libra RCNN [78]	44.68	27.08	49.27	55.21
FCOS [11]	17.90	2.88	12.95	31.15
FreeAnchor [79]	44.26	25.99	49.37	55.34
RetinaNet-SM [44]	48.48	29.01	54.28	59.95
Faster RCNN-SM [44]	51.33	33.91	55.16	62.58
RetinaNet-S-α [48]	48.34	28.61	54.59	59.38
Faster RCNN-S-α [48]	48.39	31.68	52.20	60.01
FSANet (Ours)	53.52	37.65	58.34	63.89

swimming-pool), which strongly proves the effectiveness of our approach. Qualitative detection results are visualized in Fig. 13. It is observed that FSANet is capable of not only generating precise bounding boxes for tiny objects but also working robustly under various challenging scenes. Our method can successfully locate objects and obtain satisfactory accuracy, whether facing scenes with densely arranged objects or scenes with low contrast between objects and backgrounds.

2) Results on GF1-LRSD: Table X presents the results achieved by our FSANet and other methods on GF1-LRSD test set, including Faster-RCNN [8], YOLOv5 [73], FCOS [11], CornerNet [76], R^3Det [16], and LR-TSDet [40]. Obviously, the proposed FSANet achieves the best detection performance of all other methods and outperforms the state-of-the-art of LR-TSDet by 3.54%, reaching 87.41% mAP. The mAP is calculated under VOC2012 metric [65]. The anchor-based detectors, such as R^3Det and SCRDet, require careful designs of the numbers and aspect ratios of anchor boxes to match tiny objects. But our method gets rid of the constraints of anchors, which predicts bounding boxes in a pixel-by-pixel fashion. Therefore, it yields an improvement by 14.37% mAP compared to R^3Det. Furthermore, we visualize the qualitative

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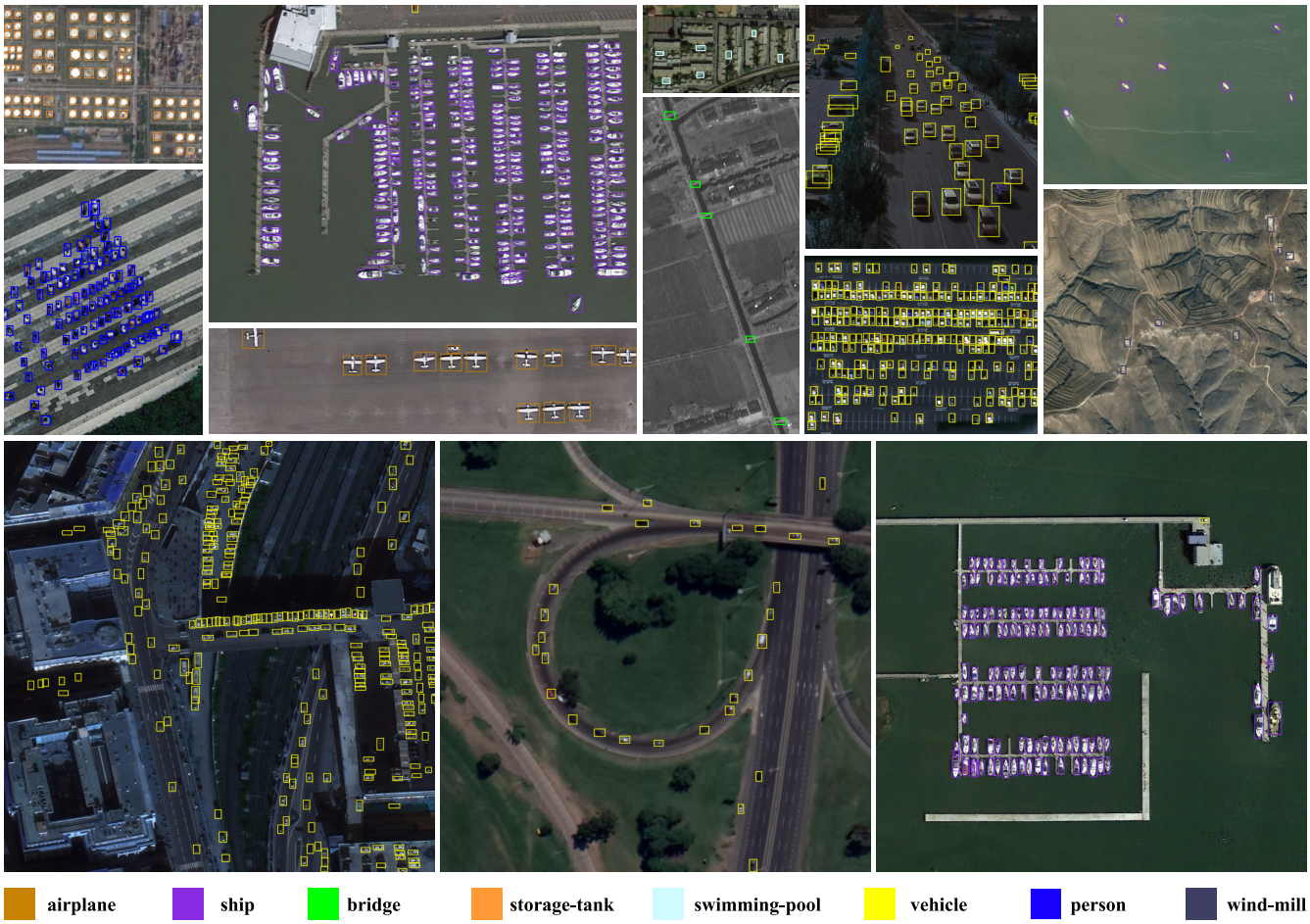


Fig. 13. Detection results achieved by our FSANet on AI-TOD test set. One color stands for one object class. It can be seen that FSANet could generate precise bounding boxes for tiny objects, and work well in complex and dense scenes. (Best viewed in color and with zoom.)

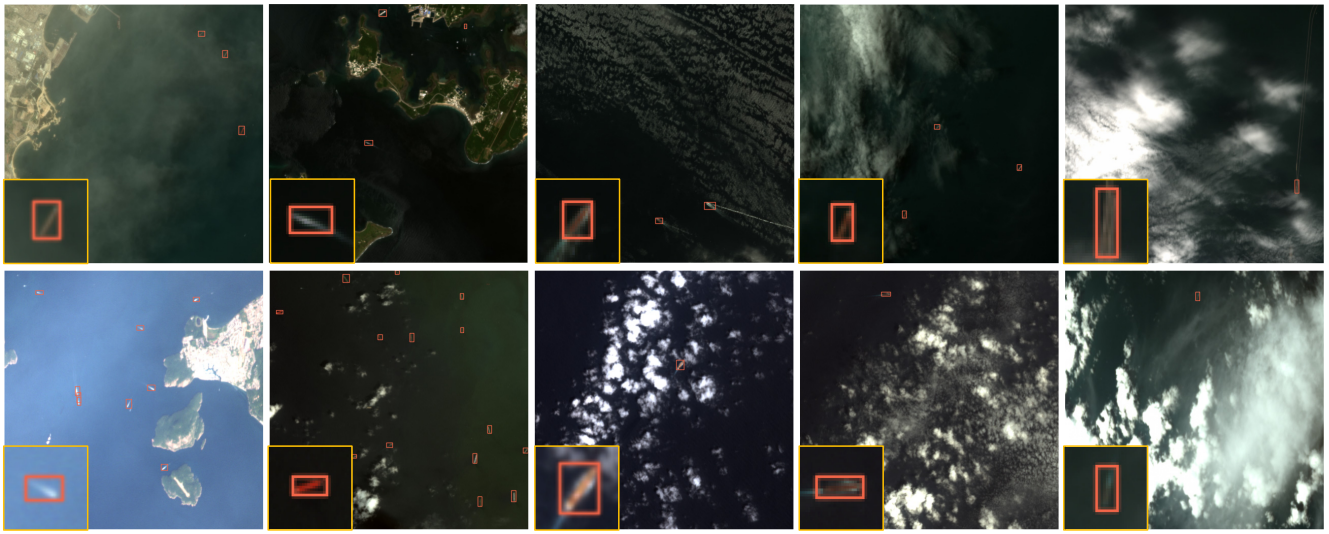


Fig. 14. Detection results achieved by our FSANet on GF1-LRSD test set. The bottom-left corner of the image shows details of zoomed-in results. FSANet could handle different real-world scenarios of tiny ship detection, including occlusion, interference, etc. (Best viewed in color and with zoom.)

detection results under different real-world circumstances in Fig. 14. The characteristics of tiny ships are not discriminative enough, and our FSANet can suppress false alarms caused by the background inference, including reefs and flocculent clouds. For occlusion by thin clouds, our detector also shows excellent performance.

3) Results on TinyPerson: We compared the detection performance achieved by our FSANet with other previous methods on TinyPerson, including RetinaNet [9], Faster RCNN [8], Grid RCNN [77], Scale Match (SM). [44], and S-α [48]. Table XI lists the detailed comparison results of objects in different size ranges under the AP_{50} metric. It can be found that

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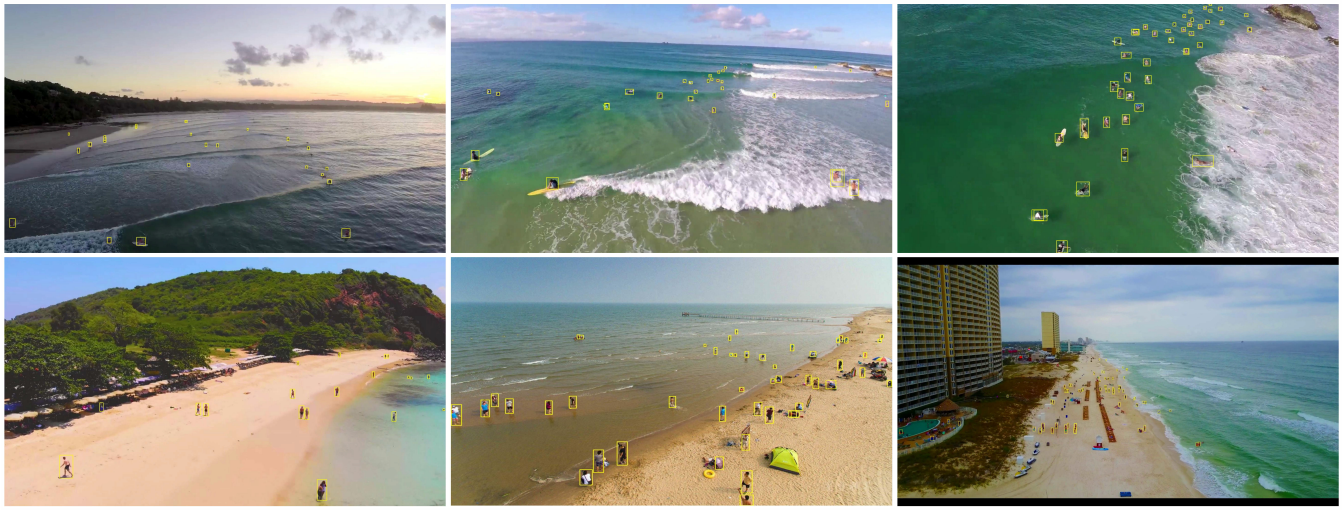


Fig. 15. Detection results achieved by our FSANet on TinyPerson test set.

our FSANet outperforms other competitors by an obvious margin, achieving the advanced performance of 53.52% AP_{50}^{tiny}. The proposed model achieves significant improvements in other metrics as well. Particularly, our FSANet obtains 3.74% AP_{50}^{tiny1}, 3.16% AP_{50}^{tiny2}, and 1.31% AP_{50}^{tiny3} improvements over Faster RCNN with SM [44], respectively. In addition, the prediction results for TinyPerson dataset are depicted in Fig. 15. As can be seen, objects under different scenarios can be accurately located, further demonstrating the effectiveness of our method.

V. Conclusion

In this article, we propose a novel anchor-free detector, namely FSANet, to improve the performance of tiny object detection in RSIs. In particular, two effective modules are introduced to address the existing challenges. We present a FA^2M to handle the feature misalignment during fusing different-resolution features. A SAGH is proposed to improve the localization capability of the network for tiny objects. Furthermore, extensive experiments show that our FSANet significantly outperforms other methods on three benchmarks, demonstrating the superiority and robustness of the proposed approach. In future work, we will focus on transferring the model knowledge to build a lightweight network that meets real-time demands while keeping competitive performance.

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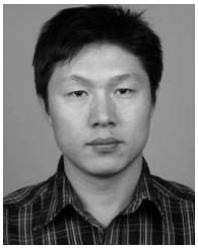
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